Surya Vardhan Senior Embedded Engineer

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Senior Embedded Firmware Engineer with 7+ years of experience delivering production firmware and system-level embedded software. Experienced in owning modules end-to-end, driving release readiness, debugging optimization issues, and integrating drivers and communication stacks across MCU + Embedded Linux platforms.

Core Skills

Languages: Embedded C (Advanced), C++ (Basic), Python, Shell Scripting, bare-metal coding.
RTOS & OS: FreeRTOS, TI-RTOS, Micrium RTOS, Embedded Linux, Device Tree (basic), Multithreading, IPC (DBus)
Embedded: Device Drivers, Board Bring-up, DMA, Interrupts, Boot Flow, Static Analysis, Unit Testing, BSP
Protocols: SPI, I2C, UART/USART, CAN, TCP/IP (basic), RS232
MCU/SoC: STM32 (M0+/M3/M4/M7), ESP32/ESP8266, TI AM263x, AM68/AM69 (A57 + M4)
Debug Tools: JTAG/SWD, Logic Analyzer, Oscilloscope, GDB, CANoe/CANalyzer, TRACE32 (basic)
Dev Tools: STM32CubeIDE, IAR, Keil, TI Code Composer Studio, ESP-IDF, VS Code, Git, Jira, Bitbucket, Confluence

Work Experience

Senior Embedded Engineer — Dec 2023 — Present
Bosch Global Software Technologies Pvt Ltd | Bengaluru

- Developed production-ready firmware and embedded RTOS applications focused on next-generation home appliance electronics under the Bosch Home Appliances, ensuring seamless integration.
- Developed a robust DBus-based communication layer that enabled efficient request/response messaging and asynchronous notifications, streamlining interactions between 10+ system components and reducing latency by 20% across the application.
- Built STM32G0B1 (ARM Cortex-M0+) firmware using **Micrium RTOS**, including real-time state machine control and hardware variant handling. Owned module-level delivery including design reviews, debugging, and release readiness for production builds.
- Resolved critical release build failures by fixing **25+ shared-variable concurrency issues** (volatile audit), enabling stable optimization with **armclang (-Os)**.
- Designed and integrated **overcurrent protection (OCP)** with interrupt-driven fault detection and emergency shutdown callbacks to prevent hardware damage.
- Integrated low-level drivers: **ADC current sensing, PWM motor control, SPI, DMA, GPIO protocols, WS2812 LEDs**, including calibration and error handling.
- Developed **SoC/SoH battery monitoring logic** and implemented unit tests using **pytest** to improve regression coverage.

Senior Engineer – Software Design — Apr 2022 — Dec 2023
Mistral Solutions Pvt Ltd | Bengaluru

- Executed full SDLC including requirements, design, development, unit testing, static analysis, debugging, maintenance, and releases.
- Ported MMCSD device driver for TI Sitara AM263x using TI Code Composer Studio and delivered within 3 months.
- Improved Jacinto (AM68/AM69) Linux test farm integration by 25% for automation testing on development boards.
- Created board-specific boot binaries and supported Linux bring-up using device tree integration.
- Collaborated in Agile/Scrum environment and contributed to technical recruitment efforts.

Embedded Software Engineer July 2021 — Feb 2022
Nash Tech Labs | Bengaluru

- Led a team of 4+ engineers for hardware/software integration for EV on-board charger development.
- Developed 32-bit MCU firmware for a 3.3 kW EV charging station using IAR and STM32CubeIDE.
- Performed circuit tracing, system requirement analysis, and firmware debugging using logs and test reports.
- Authored technical documentation: SRS, design descriptions, verification reports, datasheets, and PCB verification.

Senior Embedded Engineer Aug 2019 — Jun 2021
Senpronics | Bengaluru

- Developed bare-metal code compliant with MISRA-C guidelines for security and industrial products.
- Improved system performance and stability using multithreading and IPC concepts in IoT-based products.
- Reduced development cost by 25% by building Windows tooling (LabVIEW-based) for MCU communication over serial protocol.
- Delivered mission-critical defense products using FreeRTOS with IPC and multithreading concepts.
- Worked on high-bandwidth Wi-Fi data transmission for DAQ systems and RF signal generator firmware improvements.
- Developed low level driver development for MCU peripherals (SPI/I2C/UART/GPIO) and integrated with FreeRTOS-based application layers.

Certification

Advanced Embedded Systems – Vector India, Chennai Jul 2019
Udemy – Prompt Engineering for AI

Education

BE in Mechatronics Engineering May 2013 — April 2017
Anna University (Affiliated)
Coimbatore Institute of Engineering and Technology

Patent

Automated Security System and Method for Motor Bike Locking. Issue: Sept 2016
Application number:20164103054
International Classification: G08B21/00

Awards

Pride Pack (Outstanding Performance) – Mistral Solutions (AM68 & AM69 SK SDK release delivery).

Languages

English (Fluent) | Hindi (Fluent) | Tamil (Native) | Kannada (Conversational) | Telugu (Conversational) | German (A1)

Projects

- AM64x IPC R5 to A53 with ADC
- Hi-Speed data transfer over TCP on ESP32
- Velocity Tracker
- RF Synthesizer

Products

- Earthquake Detector [EMD - 1202®] | Seismic DAQ [PS-DAQ®] | Real Time Acceleration Plotter [RTAP-335®]